

Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

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High-dimensional embeddings support more semantic directions than can be made exactly orthogonal, but useful quasi-dimensionality occupies an intermediate regime. If $N \leq d_{\text{eff}}$, an exact orthogonal code remains geometrically available; if N is too large, pairwise crowding violates a chosen overlap tolerance, and if too many directions k are jointly active, their residual overlaps accumulate at readout. For a tolerated overlap ε , failure probability δ , active-set size k , and readout-noise tolerance τ , we derive an operational Goldilocks bracket

$$\max\{d_{\text{pack}}(N, \varepsilon, \delta), k/\tau^2\} \lesssim d_{\text{eff}} < N.$$

The packing term is computable from the exact spherical-cap probability; the second term follows from the interference scale $\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}$. This separates *kinetic capacity*—what the geometry can host—from *epistemic accessibility*—what a finite readout can recover. On this foundation we give a quantum-inspired but classically implementable model of polysemy: context observables share a token-associated spectral projector while acting differently, and generically noncommutatively, on sense subspaces in its orthogonal complement. Conditional projection onto a selected observable supplies a contextual embedding. The Goldilocks bracket and the intertwined-observable construction are separately testable and require no claim of physical quantum computation.

I. INTRODUCTION

This chapter examines contextual representation in large language models (LLMs) as part of the volume’s broader inquiry into “quantum thinking.” Only a minimal orientation to LLM operation is needed here. Input text is first divided into tokens, and each token is mapped to an initial vector. Transformer blocks then use self-attention and feed-forward transformations to turn those initially context-free vectors into sequence-dependent hidden states. An output layer converts the final state into token scores, and a softmax distribution supports probabilistic next-token selection. This compact chain—tokens, embeddings, contextual transformation, and finite readout—supplies the computational background for the argument below; the details of pre-training, fine-tuning, reinforcement learning, and backpropagation are not required.

Current Transformer-based LLMs are classical computational systems. They run on classical hardware and use real-valued matrix operations, normalization, nonlinearities, copying, and generally irreversible updates. Their reliance on vectors, inner products, high dimension, or probabilistic output does not by itself make them quantum or even quantum-like. The more limited question pursued here is where ordinary high-dimensional geometry already explains the coexistence and contextual selection of semantic features, and where a stronger comparison with quantum-like formalisms might begin. Such a comparison would have to do more than rename familiar neural operations: it would need to identify a role

for contextual bases or subspaces, incompatible or non-commuting alternatives, interference, or measurement-like readout that yields explanatory or predictive value beyond a classical dynamic-embedding account.

Semantic concurrency makes this boundary concrete. A model must preserve many latent possibilities while making only a small, context-selected subset operationally accessible. Polysemy is the simplest illustration: the same token can participate in financial, river-edge, or seat/bench contexts of “bank,” yet the surrounding sequence drives it toward different contextual representations. The verb in an aircraft “banks” supplies a fourth sense, mentioned below but not used in the three-arm schematic. This relation among possibility, context, and readout connects directly to the volume’s recurring themes without implying that the underlying machine is physically quantum. It also motivates a separation between what a representation can geometrically accommodate and what a finite channel can reliably recover.

The claims below occupy different rungs of that evidential ladder. Concentration of measure and exponential packing are standard results of high-dimensional geometry. Their use to describe semantic capacity, together with the accumulated-interference estimate, is a formal model of classical representation subject to stated assumptions about effective dimension and coactivation. The reference-subspace organization is a stronger empirical hypothesis about LLM internals and is tested against classical controls. The comparison with quantum contextuality remains a limited structural correspondence; no physical quantumness, Born-rule dynamics, or semantic Kochen–Specker theorem is inferred.

A finite-dimensional activation space faces a two-sided tension. With no more than d_{eff} semantic directions,

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exact orthogonality is geometrically possible and quasi-dimensional superposition is not forced. Once $N > d_{\text{eff}}$, the code is overcomplete and approximate orthogonality becomes useful. Yet increasing N still further eventually produces too many large pairwise overlaps, while increasing the simultaneously active set k accumulates readout noise. The useful regime is therefore a parameterized Goldilocks zone between an underloaded exact-axis regime and an overcrowded interference regime.

The two sides obey different scalings. On the capacity side, concentration of measure permits exponentially many candidate semantic directions with pairwise overlap at most ε ,

$$N \lesssim \exp(c d_{\text{eff}} \varepsilon^2). \quad (1)$$

On the readout side, the same small residual overlaps add in quadrature over an active set of size k , giving an interference scale

$$\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}. \quad (2)$$

The first relation is an existential packing estimate. The second is a typical-case readout-noise estimate. Together they say that dimension is a *semantic concurrency bandwidth*: it does not mainly limit how many meanings can be geometrically stored, but how many independently recoverable semantic components can be jointly active.

Polysemy is the cleanest example. A single lexical item, such as “bank”, can access several mutually exclusive meanings, only one of which is selected in a given context. The same tension appears more generally in feature superposition, role binding, and compositional semantics [1]: many latent semantic degrees of freedom must coexist, while only a context-selected subset is read out. We use polysemy as a controlled entry point because it makes contextual selection explicit. In lexical semantics, polysemy is distinguished from homonymy by the relatedness of senses [2]. Modern Transformer-based LLMs implement contextual selection through dynamic contextual embeddings: a token representation is repeatedly transformed by self-attention and MLP layers as it absorbs information from its sentential environment [3].

The chapter develops two central ideas. First, concentration permits a large semantic dictionary, but both dictionary crowding and accumulated interference delimit a quantitative Goldilocks bracket. This geometric claim is architecture-independent once N , k , ε , τ , and d_{eff} have been operationally defined. Second, for some canonically polysemous tokens, contexts may be represented by intertwined symmetric operators. These operators share a stable token-associated spectral projector, while sense information occupies low-dimensional subspaces in its orthogonal complement. Selecting or mixing the corresponding conditional projections implements a contextual embedding. This operator construction is not implied by concentration geometry; it is a stronger empirical model, and Sec. V D states how it can fail.

A serious caveat is effective dimension. Trained representations are often anisotropic: they occupy a nar-

row cone rather than the full ambient space. The dimension controlling concentration is therefore not necessarily the raw embedding dimension d , but an effective dimension d_{eff} . If d_{eff} is small, the capacity claim weakens accordingly. For example, at $d_{\text{eff}} \sim 50$ and $\varepsilon = 0.1$, the exponent $c d_{\text{eff}} \varepsilon^2$ is only a small constant. The framework therefore applies to the approximately isotropic geometry actually used by the representation, possibly after whitening or restriction to a participation subspace. Determining which layers, models, and preprocessing regimes admit a meaningful d_{eff} is an empirical priority.

The reference-subspace and operator hypotheses have a narrower domain. They are meant for canonically polysemous tokens with established sense inventories, such as WordNet-, OntoNotes-, SemCor-, or WiC-style sense distinctions [4] with enough contextual instances for testing. Failure on a representative sample, against the controls in Sec. V D, would count against a privileged token reference direction or an intertwined-observable implementation even if the broader Goldilocks capacity/accessibility picture survives.

The proposal is adjacent to the superposition hypothesis in mechanistic interpretability [5], according to which neural networks represent more features than they have neurons by packing features into nearly orthogonal directions. The present chapter separates the geometric background from a possible semantic organization within it. The general superposition picture concerns an overcomplete dictionary of feature directions and sparse active sets. The present model adds the more specific possibility that contextual readout is organized around stable token-associated directions and context-dependent, potentially noncommuting projectors.

Section II develops the capacity framework: quasi-orthogonality, Lévy’s lemma, exponential packing, and the kinetic/epistemic distinction, culminating in the Goldilocks bracket of Sec. II E. Section III introduces the reference-subspace hypothesis, its realization by intertwined context observables, and its relation to dynamic contextual embeddings. Section IV connects the construction to attention as soft context routing rather than literal basis reconstruction. Section V discusses superposition, sparse autoencoders, limitations, empirical tests, and the limited quantum analogy.

Notation. Throughout, $\mathbb{R}^d$ is Euclidean d -space with inner product $u^\top v$ and norm $\|v\| = \sqrt{v^\top v}$; $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$ is the unit sphere; I is the identity matrix; $\|\cdot\|_{\text{op}}$ is the operator norm; $A \succeq 0$ means positive semidefinite; and $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the uniform measure on the relevant sphere unless stated otherwise. Universal positive constants are denoted c, c' , possibly changing from line to line. The symbol d_{eff} denotes the effective dimension controlling concentration in the representation being probed.

II. CONCENTRATION GEOMETRY AND CAPACITY

This section is architecture-independent. It gives the geometric capacity constraint on which the later reference-subspace hypothesis builds.

A. Quasi-orthogonality

Let $w_1, \dots, w_N \in \mathbb{R}^d$ be unit feature directions. Strict orthogonality, $w_i^\top w_j = 0$ for $i \neq j$, allows at most $N = d$ such vectors. Semantic encoding does not require exact orthogonality. It requires approximate distinguishability. We therefore call a family ε -quasi-orthogonal if

$$|w_i^\top w_j| \leq \varepsilon, \quad i \neq j. \quad (3)$$

This is the relevant notion for high-dimensional feature storage: small overlap is tolerable, provided readout can absorb the resulting interference.

B. Lévy's lemma and linear overlaps

High-dimensional spheres make quasi-orthogonality generic.

The mechanism is already visible before invoking the formal lemma. Let

$$x = (x_1, \dots, x_d) \in \mathbb{R}^d$$

be a unit vector chosen without a preferred direction. Then

$$x_1^2 + \dots + x_d^2 = 1.$$

Rotational invariance suggests that the norm is spread over the available coordinates, so a typical coordinate weight is of order $1/d$. Now fix a unit direction u . By rotational invariance one may choose coordinates so that $u = e_1$. The squared overlap with the random direction is then

$$|u^\top x|^2 = x_1^2.$$

Thus the overlap with any fixed direction is typically of order $d^{-1/2}$ in amplitude, or $1/d$ in squared amplitude. A macroscopic overlap $|u^\top x| > \varepsilon$ therefore requires one coordinate to carry an anomalously large fraction of the total norm. In high dimension such an event is exponentially unlikely.

The same intuition appears particularly transparently in the complex Hilbert-space analogue. If $\psi \in \mathbb{C}^D$ is Haar-random and ϕ is fixed, unitary invariance permits the choice $\phi = e_1$, so

$$|\langle \phi, \psi \rangle|^2 = |\psi_1|^2.$$

For a Haar-random complex unit vector the exact tail is

$$\mathbb{P}\{|\langle \phi, \psi \rangle|^2 \geq \eta\} = (1 - \eta)^{D-1} \leq \exp[-(D-1)\eta].$$

For real embedding vectors the analogous spherical concentration estimate has the operative scaling

$$\mathbb{P}\{|u^\top x| > \varepsilon\} \lesssim \exp(-cd\varepsilon^2)$$

for a universal constant $c > 0$. Consequently, if N random directions are sampled, a union-bound estimate gives a probability at most of order $N^2 \exp(-cd\varepsilon^2)$ that some pair has overlap larger than ε . Hence N may grow exponentially in $d\varepsilon^2$ while all pairwise overlaps remain small with high probability. This is the heuristic core of the exponential quasi-orthogonal capacity derived below.

Lévy's lemma makes this intuition precise. If $f : S^{d-1} \rightarrow \mathbb{R}$ is L -Lipschitz, then

$$\mathbb{P}(|f(v) - \mathbb{E}f| \geq \delta) \leq 2 \exp\left(-\frac{cd\delta^2}{L^2}\right), \quad (4)$$

for a universal $c > 0$ [6, 7]. Apply this to the linear functional $f_u(v) = u^\top v$. It is 1-Lipschitz and has mean zero by symmetry. Hence

$$\mathbb{P}(|u^\top v| \geq \varepsilon) \leq 2 \exp(-cd\varepsilon^2). \quad (5)$$

For normalized surface measure one obtains the explicit form $2 \exp(-(d-1)\varepsilon^2/2)$ [6]. Thus two random unit vectors are nearly orthogonal with high probability, and typical overlap amplitudes scale as $d^{-1/2}$.

In trained models, d should be replaced by d_{eff} , the dimension actually controlling concentration. Anisotropy, low intrinsic dimension, or a narrow activation cone reduce d_{eff} . Whitening or projection to a participation subspace may increase the effective dimension relevant for the analysis.

Relation to Johnson–Lindenstrauss. The Johnson–Lindenstrauss lemma preserves pairwise distances among n points after projection to dimension $k = O(\varepsilon^{-2} \log n)$ [8]. It is driven by the same concentration phenomenon as Eq. (5); indeed JL can be derived from sphere concentration applied to projection norms [9]. Here Lévy's lemma is the more direct tool: the issue is not projecting a finite dataset, but the typical overlap between semantic directions in the representation space.

C. Exponential packing

Sample N unit vectors independently and uniformly from S^{d-1} . By Eq. (5) and a union bound over pairs,

$$\mathbb{P}(\exists i \neq j : |w_i^\top w_j| \geq \varepsilon) < N^2 \exp(-cd_{\text{eff}}\varepsilon^2). \quad (6)$$

This probability is below one, and hence an ε -quasi-orthogonal family exists, provided

$$N < \exp\left(\frac{c}{2} d_{\text{eff}} \varepsilon^2\right). \quad (7)$$

This is the kinetic capacity estimate: in high effective dimension, many candidate semantic directions can coexist with small pairwise overlap.

The estimate controls pairwise overlap only. It does not guarantee that a large collection of feature directions is well-conditioned. By Gershgorin, $|w_i^\top w_j| \leq \varepsilon$ gives only $\|G - I\|_{\text{op}} \leq (n-1)\varepsilon$ for an n -element Gram matrix, which is useless when n is large. Joint readout therefore needs a separate conditioning assumption, introduced in Sec. III C.

D. Kinetic capacity versus epistemic accessibility

The packing bound says what can coexist. It does not say what can be read. We call the coexistence supply *kinetic semantic capacity*. We call the recoverable subset *epistemic semantic accessibility*. The distinction is the central point of the chapter.

Let N candidate features be represented by unit directions $w_1, \dots, w_N \in \mathbb{R}^{d_{\text{eff}}}$. In a given context, suppose an active set S of size k contributes amplitudes x_j , producing

$$h = \sum_{j \in S} x_j w_j. \quad (8)$$

A linear readout of feature i gives

$$w_i^\top h = x_i + \sum_{\substack{j \in S \\ j \neq i}} x_j w_i^\top w_j. \quad (9)$$

The first term is signal; the second is interference. For random incoherent directions,

$$\mathbb{E}[w_i^\top w_j] = 0, \quad \text{Var}(w_i^\top w_j) \simeq \frac{1}{d_{\text{eff}}}. \quad (10)$$

Thus, conditional on the active amplitudes,

$$\text{Var}\left(\sum_{\substack{j \in S \\ j \neq i}} x_j w_i^\top w_j\right) \simeq \frac{1}{d_{\text{eff}}} \sum_{\substack{j \in S \\ j \neq i}} x_j^2. \quad (11)$$

For amplitudes of order unity,

$$\sigma_{\text{int}} \sim \sqrt{\frac{k}{d_{\text{eff}}}}. \quad (12)$$

This is the readout bottleneck. Concentration lets many directions coexist because their overlaps are small. But readout sees the *sum* of many small overlaps. Reliable recovery requires

$$\sqrt{\frac{k}{d_{\text{eff}}}} \ll \Delta, \quad (13)$$

where Δ is the relevant decision margin. Equivalently,

$$\text{kinetic: } N \lesssim \exp(c d_{\text{eff}} \varepsilon^2), \quad \text{epistemic: } k \lesssim d_{\text{eff}} \Delta^2. \quad (14)$$

The first is storage-like coexistence. The second is simultaneous accessibility. A model may host many more features than it can jointly read.

E. A Goldilocks bracket for useful quasi-dimensionality

The preceding scalings delimit an intermediate regime, but an estimate of that regime requires the quantities being held fixed to be stated. We use N for the number of represented semantic directions and d_{eff} for the effective carrier dimension. Calling either one simply “the number of dimensions” obscures the bracket.

For two independent real unit vectors in $\mathbb{R}^d$, the exact probability of an overlap larger than ε is

$$p_d(\varepsilon) = \mathbb{P}\{|u^\top v| > \varepsilon\} = I_{1-\varepsilon^2}\left(\frac{d-1}{2}, \frac{1}{2}\right), \quad (15)$$

where $I_x(a, b)$ is the regularized incomplete beta function. A union bound for an isotropic random dictionary gives

$$\mathbb{P}\left\{\max_{i < j} |w_i^\top w_j| > \varepsilon\right\} \leq \binom{N}{2} p_{d_{\text{eff}}}(\varepsilon). \quad (16)$$

For an integer isotropic carrier, Eq. (15) is exact. When d_{eff} is a noninteger participation-ratio estimate, using $p_{d_{\text{eff}}}$ is an interpolation that should be calibrated against matched random controls rather than interpreted as an exact sphere identity.

For a chosen failure budget δ , define the random-code design ceiling

$$N_{\text{typ}}(d, \varepsilon, \delta) = \left\lfloor \frac{1 + \sqrt{1 + 8\delta/p_d(\varepsilon)}}{2} \right\rfloor. \quad (17)$$

At fixed d_{eff} , the quasi-dimensional Goldilocks band is therefore

$$d_{\text{eff}} < N \leq N_{\text{typ}}(d_{\text{eff}}, \varepsilon, \delta). \quad (18)$$

Below the left boundary, $N \leq d_{\text{eff}}$, exact orthogonal axes are geometrically available. These may be called “viewer-independent” only in the restricted sense that one global orthogonal coordinate system can label all directions. This does not make semantics itself context-independent; it only means that overcompleteness is not forced by dimension. Above the right boundary the high-probability guarantee for an isotropic random code is lost and excessive overlaps become typical under the null model.

The same band can be written as a bracket on dimension. For fixed N , let

$$d_{\text{pack}}(N, \varepsilon, \delta) = \min \left\{ d : \binom{N}{2} p_d(\varepsilon) \leq \delta \right\}. \quad (19)$$

If τ is the largest acceptable rms interference, Eq. (12) also requires

$$d_{\text{read}}(k, \tau) = \frac{k}{\tau^2}. \quad (20)$$

Thus an operational dimension bracket is

$$d_{\min} = \max\{d_{\text{pack}}(N, \varepsilon, \delta), d_{\text{read}}(k, \tau)\} \lesssim d_{\text{eff}} < N. \quad (21)$$

Using the exponential cap bound and ignoring polynomial prefactors gives the transparent estimate

$$d_{\text{pack}} \approx 1 + \frac{4 \log N + 2 \log(1/\delta)}{\varepsilon^2}. \quad (22)$$

The exact cap probability in Eq. (15) is preferable for numerical work. For example, $N = 10^4$, $\varepsilon = 0.10$, and $\delta = 0.05$ give $d_{\text{pack}} \simeq 3716$. If $k = 64$ components are jointly active and $\tau = 0.125$, then $d_{\text{read}} = 4096$, so the illustrative band is

$$4.1 \times 10^3 \lesssim d_{\text{eff}} < 10^4. \quad (23)$$

Equivalently, at $d_{\text{eff}} = 4096$ the same cap calculation gives $N_{\text{typ}} \simeq 2.7 \times 10^4$, hence $4096 < N \lesssim 2.7 \times 10^4$ for this particular overlap and reliability criterion.

This is an estimate, not a universal numerical constant or a sharp upper bound for optimized spherical codes. Learned directions are not independent Haar samples; coactivation is structured; nonlinear readout may tolerate larger overlaps; and d_{eff} varies by layer and preprocessing. The bracket is useful precisely because all of these choices are exposed as measurable parameters. If $d_{\min} \geq N$, the quasi-dimensional Goldilocks interval is empty for the stated tolerances.

Figure 1 gives a minimal synthetic illustration of this distinction. The simulation is not intended as a model of a trained language model. It is only a null model for the concentration geometry: random unit directions in an effective dimension d_{eff} can coexist in large quasi-orthogonal families, while simultaneous linear readout accumulates the residual overlaps of the active directions. Deviations from this null model in trained representations would themselves be informative, reflecting anisotropy, learned feature alignment, nonlinear readout, or coactivation-dependent orthogonalization.

This also clarifies the relation to compressed sensing. Stable recovery of a k -sparse vector in an N -dimensional feature dictionary from m measurements typically requires $m \gtrsim Ck \log(N/k)$ under suitable incoherence or restricted-isometry assumptions [10–12]. In an idealized isotropic residual stream, one may identify m with d_{eff} ; in trained models the effective representation dimension and effective readout dimension may differ. The compressed-sensing analogy is therefore diagnostic, not literal.

III. TOKEN REFERENCE SUBSPACES AND INTERTWINED CONTEXT OBSERVABLES

The previous section established an architecture-independent capacity constraint. We now ask a more speculative question: does a trained model organize the

contextual states of some polysemous tokens relative to a stable token-associated reference subspace? This hypothesis is not implied by concentration geometry, and it is separately falsifiable.

The notation distinguishes mathematical types and roles. For a token t and a sense context C :

- $r_t \in \mathbb{R}^d$ is a unit candidate token reference vector;
- $\mathcal{R}_t \subset \mathbb{R}^d$ is the token reference subspace;
- P_t is the orthogonal projector onto $\mathcal{R}_t$;
- $S_{t,C} \subset \mathcal{R}_t^\perp$ is a low-dimensional sense subspace;
- $u_{t,C,j} \in S_{t,C}$ are vectors spanning that sense subspace;
- $h_\ell(t \mid x) \in \mathbb{R}^d$ is an observed contextual hidden state.

Thus r_t , $u_{t,C,j}$, and $h_\ell(t \mid x)$ are vectors, whereas $\mathcal{R}_t$ and $S_{t,C}$ are subspaces. The w_i used in Sec. II have a separate role: they are generic feature directions in the capacity calculation and are not reused as sense labels.

A. The reference-subspace hypothesis

In the simplest version, let r_t be a candidate unit direction and set

$$\mathcal{R}_t = \text{span}\{r_t\}. \quad (24)$$

Empirically, r_t could be the normalized static input embedding, a normalized mean contextual embedding, or a learned token-specific probe direction. It is simply a hypothesized context-stable reference for token identity. It is not a mechanical or dynamical object. Any state can be decomposed relative to any direction, so the decomposition alone has no empirical content. The claim is that a token-associated $\mathcal{R}_t$ exposes sense structure better than matched control directions or subspaces.

For each context or sense label C , define a low-dimensional subspace

$$S_{t,C} = \text{span}\{u_{t,C,1}, \dots, u_{t,C,q_C}\} \subset \mathcal{R}_t^\perp, \quad q_C \ll d_{\text{eff}}. \quad (25)$$

For the intended three-arm example, finance, river-edge, and seat/bench associations of “bank” correspond to different subspaces $S_{\text{bank},C}$ inside the common orthogonal carrier $\mathcal{R}_{\text{bank}}^\perp$. The verbal use in which an aircraft banks or tilts supplies a fourth sense and would add a fourth context subspace; it is mentioned here but omitted from the drawing for clarity. A word sense is not a single vector: topic, syntax, selectional preferences, and world knowledge may require several spanning directions $u_{t,C,j}$.

The reference object may itself require more than one direction. In that case define

$$\mathcal{R}_t = \text{span}\{r_{t,1}, \dots, r_{t,p}\}, \quad p \geq 1, \quad (26)$$

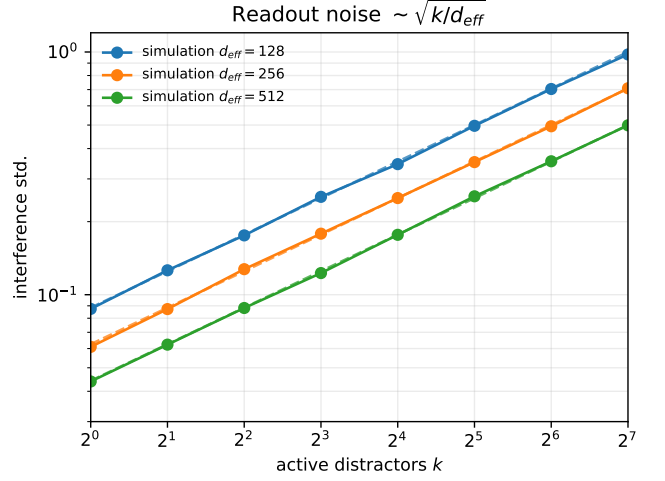
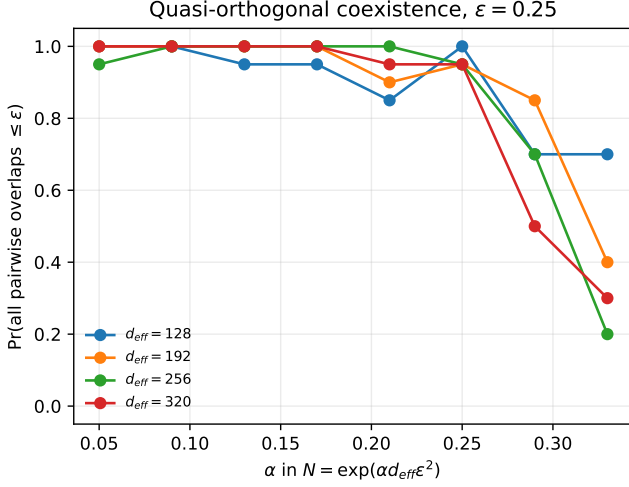


FIG. 1. Synthetic geometry checks for the capacity/accessibility distinction. Left: random unit directions in dimension d_{eff} are sampled with $N = \exp(\alpha d_{\text{eff}} \varepsilon^2)$ and tested for ε -quasi-orthogonality. For sufficiently small α , large families coexist with high probability, illustrating the exponential packing side of Eq. (14). Right: for a fixed target direction, k random distractor directions are simultaneously active with random signs. The standard deviation of the readout interference follows the predicted scale $\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}$, illustrating the accessibility side of Eq. (14). Together the panels visualize the two lower constraints entering Eq. (21). Both use random isotropic vectors and should be read as a geometric null model rather than as measurements on a trained language model.

and require

$$S_{t,C} \subset \mathcal{R}_t^\perp. \quad (27)$$

Equation (24) is the special case $p = 1$. The bare subspace construction specifies only a pattern of sharing. In quantum logic, different contexts can share one or more observables or spectral projections; finite-dimensional examples with two shared observables are known [13]. In the semantic model, $\mathcal{R}_t$ is a common token-associated reference subspace, while the sense-specific subspaces lie in its orthogonal complement. These incidence relations alone imply neither noncommutativity nor quantum probability. The next subsection makes the stronger, optional operator structure explicit.

The low-rank assumption is empirical. It predicts that dominant within-sense variation is low-dimensional after projection onto $\mathcal{R}_t^\perp$. This is compatible with evidence that contextualized embeddings can separate word senses, but it is stronger: it asks whether such separation is privileged relative to a token-associated reference subspace [14, 15].

B. Intertwined context observables and conditional embeddings

The shared-subspace idea can be implemented as a family of intertwined context observables. Let $Q_t \in \mathbb{R}^{d \times p}$ have orthonormal columns spanning $\mathcal{R}_t$, so that $P_t = Q_t Q_t^\top$. For each context C , let $B_{t,C} \in \mathbb{R}^{d \times q_C}$ have orthonormal columns spanning $S_{t,C}$. Here Q_t and $B_{t,C}$ are basis matrices, not semantic vectors.

The projector onto the token-plus-sense subspace is

$$\Pi_{t,C} = P_t + B_{t,C} B_{t,C}^\top. \quad (28)$$

All $\Pi_{t,C}$ contain the same spectral projector P_t and therefore intertwine on $\mathcal{R}_t$. Their sense parts differ. In particular,

$$[\Pi_{t,C}, \Pi_{t,C'}] = [B_{t,C} B_{t,C}^\top, B_{t,C'} B_{t,C'}^\top] \quad (29)$$

need not vanish. Nonzero commutators represent incompatible conditional decompositions: applying the two context projections in opposite orders can give different intermediate states.

A symmetric context observable with this eigenspace structure is, for example,

$$A_{t,C} = Q_t \Lambda_t Q_t^\top + B_{t,C} \Lambda_{t,C} B_{t,C}^\top, \quad (30)$$

where the diagonal matrix Λ_t is shared across contexts and $\Lambda_{t,C}$ is context-specific; their eigenvalue sets may be chosen disjoint so that P_t is identifiable as a shared spectral sector. The operator acts as zero on the unused orthogonal complement; an orthonormal completion may assign additional eigenvalues without changing the shared structure. With a nondegenerate completed spectrum, $A_{t,C}$ is the real finite-dimensional analogue of a maximal observable. Generic choices for different C need not commute, although every $A_{t,C}$ retains the common spectral sector P_t .

The corresponding context-conditioned embedding is concrete:

$$h_{t,C}(x) = \Pi_{t,C} h(x) = P_t h(x) + B_{t,C} B_{t,C}^\top (I - P_t) h(x). \quad (31)$$

One may select a single C , or take a soft mixture of these embeddings as in Sec. IV. Equations (28)–(31) provide the proposed implementation of intertwining observables for contextual embedding. They use ordinary real matrices and can be trained or probed in a classical network. Thus noncommutativity here is a testable representational property, not evidence of physical quantum dynamics or the Born rule.

C. Projection and gated subspace readout

Let $h(x) \in \mathbb{R}^d$ be a contextual state and retain the orthogonal projector P_t defined above. The two components are

$$\begin{aligned} h(x) &= h_{\text{ref}}(x) + h_{\perp}(x), \\ h_{\text{ref}}(x) &= P_t h(x), \\ h_{\perp}(x) &= (I - P_t)h(x) \in \mathcal{R}_t^{\perp}. \end{aligned} \quad (32)$$

For the one-dimensional case, $P_t = r_t r_t^T$ and the scalar $a_t(x) = r_t^T h(x)$ measures the component along the reference direction. The hypothesis is that most sense-disambiguating variation is present in $h_{\perp}(x)$, not that the projection formula itself is special.

Readout inside $S_{t,C}$ also requires the spanning vectors to be well-conditioned. Let

$$U_{t,C} = [u_{t,C,1} \cdots u_{t,C,q_C}], \quad G_{t,C} = U_{t,C}^T U_{t,C}. \quad (33)$$

The columns of $U_{t,C}$ span $S_{t,C}$; $U_{t,C}$ is a matrix, not an additional semantic vector. Unlike the orthonormal $B_{t,C}$ used in the spectral construction, $U_{t,C}$ may be a learned nonorthogonal spanning matrix. We call it ρ -well-conditioned if

$$\|G_{t,C} - I\|_{\text{op}} \leq \rho < 1. \quad (34)$$

Then

$$\tilde{U}_{t,C} = U_{t,C} G_{t,C}^{-1} \quad (35)$$

is a dual spanning matrix and satisfies $\|\tilde{U}_{t,C}\|_{\text{op}} \leq (1 - \rho)^{-1/2}$. Near-degenerate spanning sets amplify readout noise, so the model requires ρ comfortably below one. The corresponding coordinates are

$$\alpha_{t,C}(x) = \tilde{U}_{t,C}^T h_{\perp}(x) \in \mathbb{R}^{q_C}. \quad (36)$$

A simple sense score is

$$s_{t,C}(x) = g_t(a_t(x)) f_{t,C}(h_{\perp}(x)), \quad (37)$$

where $g_t \geq 0$ gates engagement with the token reference direction and $f_{t,C} \geq 0$ measures fit to the sense subspace. One possible choice is $f_{t,C} = \|\alpha_{t,C}(x)\|^2$; a learned positive quadratic form is another. The selected sense label is

$$C^*(x) = \arg \max_C s_{t,C}(x). \quad (38)$$

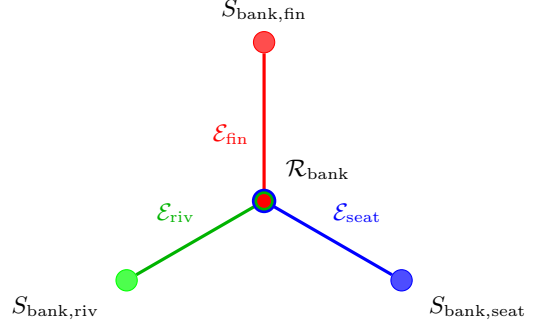


FIG. 2. Hypergraph schematic of context organization for the polysemous token “bank.” The original star geometry is retained, but its elements are now typed as abstract hypergraph objects rather than spatial vectors. The central vertex denotes the shared token reference subspace $\mathcal{R}_{\text{bank}}$; the colored outer vertices denote the complete low-dimensional sense subspaces $S_{\text{bank},C} \subset \mathcal{R}_{\text{bank}}^{\perp}$, not individual directions. Each colored spoke is the context hyperedge $\mathcal{E}_C = \{\mathcal{R}_{\text{bank}}, S_{\text{bank},C}\}$ for $C \in \{\text{fin}, \text{riv}, \text{seat}\}$. Each hyperedge is the combinatorial shadow of a context projector $\Pi_{\text{bank},C}$ or observable $A_{\text{bank},C}$. Because each displayed hyperedge has two abstract vertices, it is drawn as an ordinary line; a context with additional components would be a higher-order hyperedge incident on all of them. The star is an incidence diagram, not a coordinate plot, and its spokes do not denote vector directions, maps, or causal flow.

The empirical content is the proposed separation of roles: the reference component tracks token identity or engagement, while the orthogonal component contains sense-disambiguating information. If no token-associated $\mathcal{R}_t$ exposes sense structure better than matched random, frequency-controlled, or layer-matched subspaces, the hypothesis fails.

Figure 2 illustrates these objects as an abstract hypergraph incidence structure.

D. Packing sense subspaces in a shared orthogonal carrier

The vector packing bound extends to low-dimensional subspaces. Fix a unit reference vector $r_0 \in S^{d-1}$ and let $m = d - 1$. For $1 \leq q \ll m$ and any fixed threshold $\mu > 2\sqrt{q/m}$, there exist q -dimensional subspaces $S_1, \dots, S_M \subset r_0^{\perp}$ such that

$$\|U_a^T U_b\|_{\text{op}} \leq \mu, \quad a \neq b, \quad (39)$$

where $U_a, U_b \in \mathbb{R}^{m \times q}$ are orthonormal basis matrices, and

$$M \geq \exp(c_{\mu,q} m) \quad (40)$$

for some positive $c_{\mu,q}$.

This is a standard Grassmannian packing consequence of concentration on Stiefel and Grassmann manifolds [16,

[17]. For two independent Haar-random q -planes, the singular values of $U^T V$ are the cosines of their principal angles. In the regime $q \ll m$, the largest principal correlation is of order $\sqrt{q/m}$ [18–20]. A union bound over sampled subspaces then gives an exponential packing.

For trained representations this statement must be read inside an approximately isotropic effective carrier. If $\dim(\mathcal{R}_t) = p$, its effective dimension is $m_{\text{eff}} = d_{\text{eff}} - p$, not necessarily the raw ambient dimension minus p . Without such an effective carrier, the Haar-random subspace model is not a good description of activation geometry.

E. Relation to dynamic contextual embeddings

Transformers already handle polysemy dynamically. The hidden state $h_\ell(t | x)$ of token t at layer ℓ depends on context x . The reference-subspace and intertwined-observable hypotheses are not alternative hardware architectures; they are proposed organizations and read-outs of these observed trajectories.

Given a candidate reference subspace $\mathcal{R}_t$ and its projector P_t , write

$$\begin{aligned} h_\ell(t | x) &= P_t h_\ell(t | x) + z_\ell(t, x), \\ z_\ell(t, x) &= (I - P_t) h_\ell(t | x) \in \mathcal{R}_t^\perp. \end{aligned} \quad (41)$$

The decomposition is automatic; the hypothesis is not. It predicts that $P_t h_\ell(t | x)$ tracks token identity or engagement, while $z_\ell(t, x)$ carries most sense-disambiguating information. Sense labels should be more separable in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces. For each sense C , the projected states $z_\ell(t, x)$ should also concentrate near the low-dimensional subspace $S_{t,C} \subset \mathcal{R}_t^\perp$.

Thus the critical question is not whether contextual embeddings move; they do. It is whether their sense-relevant movement has a privileged organization relative to a token-associated reference subspace, and whether conditional application of $\Pi_{t,C}$ explains that movement better than matched unstructured projectors.

IV. ATTENTION AND LATENT CONTEXT SELECTION

A standard attention head computes

$$h' = \sum_j \alpha_j \nu_j, \quad \alpha_j = \frac{\exp(q^T k_j / \sqrt{d})}{\sum_{j'} \exp(q^T k_{j'} / \sqrt{d})}. \quad (42)$$

Keys and values are produced by different learned maps, so value vectors are not generally coefficients of a dual basis for the key vectors. Attention is therefore not literal basis reconstruction.

The useful analogy is softer. Attention routes information: it amplifies some contextual directions and suppresses others. In the terminology of Sec. III, this resembles context-dependent subspace selection. A literal

basis-reconstruction interpretation would require additional constraints, such as approximately Parseval keys and values implementing the corresponding dual reconstruction. We do not assume this.

A compact latent-variable version is

$$p(y | x, t) = \sum_C p(y | x, t, C) p(C | x, t), \quad (43)$$

where t is the token being disambiguated, y is a possible output token, and C indexes candidate sense contexts. One may model

$$p(C | x, t) = \frac{\exp(s_{t,C}(x)/T)}{\sum_{C'} \exp(s_{t,C'}(x)/T)}, \quad (44)$$

with $s_{t,C}$ the gated score of Eq. (37). This is not a claim about the literal implementation of decoding in current LLMs: standard temperature acts on token logits, not on explicit latent sense labels. The point is only that ambiguity can be represented as a distribution over possible sense contexts rather than as a single fixed interpretation.

The same weights provide a soft implementation of the intertwined-observable construction:

$$\bar{h}_t(x) = \sum_C p(C | x, t) h_{t,C}(x) = \sum_C p(C | x, t) \Pi_{t,C} h(x). \quad (45)$$

Hard selection uses C^* from Eq. (38); soft selection uses Eq. (45). Attention may supply the routing signal $p(C | x, t)$, but the structured projectors or observables are an additional model component and should not be identified with standard attention weights without empirical evidence.

V. DISCUSSION

The geometric proposal is the capacity/accessibility distinction sharpened into the Goldilocks bracket (21). The intertwined-observable model is one possible contextual organization within that bracket, not a consequence of concentration geometry.

A. Relation to superposition and sparse autoencoders

The superposition hypothesis holds that neural networks represent more features than they have neurons by packing features into nearly orthogonal directions [5, 21]. The concentration estimate (5) gives the kinetic side of that picture: many directions can coexist. Equation (12) gives the epistemic side: many jointly active features create interference of scale $\sqrt{k/d_{\text{eff}}}$. Sparsity is therefore not an implementation detail; it is necessary for readout.

Sparse autoencoders (SAEs) give a concrete empirical interface. They approximate layer activations as sparse expansions over learned decoder directions, making the w_i of Eq. (8) measurable objects rather than

TABLE I. Dynamic contextual embeddings versus the intertwined-observable hypothesis.

Dynamic description	Intertwined-observable hypothesis
Token state changes by context and layer	State is projected relative to $\mathcal{R}_t$
Layer activation carries context	$\mathcal{R}_t^\perp$ contains sense variation
Polysemy by vector deformation	Polysemy by conditional projection $\Pi_{t,C}$
Resource is depth and width	Resource is the Goldilocks bracket
Limit is activation bottleneck	Limit is crowding plus readout interference

hypothetical features. Recent work has scaled SAEs, studied SAE feature geometry, and automated interpretation of large numbers of latents [22–25]. In the present framework, SAE decoder directions can test the capacity/accessibility prediction, while sense-conditioned SAE activation clusters can test the stronger reference-subspace hypothesis.

B. A coactivation-weighted capacity limit

The active-set size k is a crude summary. A more refined description uses the coactivation graph

$$A_{ij} = \mathbb{P}(i, j \text{ active}). \quad (46)$$

If active amplitudes have, conditional on coactivation, zero mean and pairwise uncorrelated signs, then cross terms cancel in expectation and feature j contributes to the readout variance of feature i in proportion to $\mathbb{E}[x_j^2 \mid i, j \text{ active}](w_i^\top w_j)^2$. Absorbing amplitude variance and feature importance into coefficients b_i, b_j gives

$$E_{\text{int}} = \sum_{i \neq j} A_{ij} b_i b_j (w_i^\top w_j)^2. \quad (47)$$

Frequently coactive features are therefore pressured toward orthogonality. Rarely coactive or mutually exclusive features are much less constrained and may share directions or even become approximately antipodal if non-linear readout makes this useful. The relevant object is not just the number of features, but the weighted coactivation graph.

C. Limitations

The framework can fail in several ways.

First, the Goldilocks bracket is a calibrated random-code benchmark, not a model-independent hard boundary. Equation (16) assumes independent isotropic directions, while the readout boundary assumes a stated noise tolerance τ . Learned dictionaries can beat or underperform that null model. The bracket is informative only when $N, k, \varepsilon, \delta$, and τ are fixed before evaluation.

Second, anisotropy may make d_{eff} much smaller than d . Contextual embeddings are known to be anisotropic, and

intrinsic dimension can vary strongly by layer and training regime [26–29]. Whitening and related postprocessing can partially restore isotropic geometry [27, 28, 30]. At the same time, anisotropy should not be treated as an automatic or universal property of all Transformer representations [31]. The bounds should always be read in terms of the effective dimension actually achieved.

Third, the spanning matrix $U_{t,C}$ needs to be well-conditioned. If ρ in Eq. (34) is close to one, its dual amplifies readout noise.

Fourth, the reference-subspace pattern may simply be descriptively wrong. Trained models may represent polysemy without any preferred token-associated direction or subspace. In that case the capacity/accessibility account may still hold, but the reference-subspace hypothesis fails.

Fifth, the law $\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}$ assumes random-like incoherent directions and weakly correlated signs or amplitudes. Learned networks may improve on this by orthogonalizing frequently coactive features, or worsen it through anisotropy and correlated errors. The law is a scaling principle, not a universal equality.

Finally, the operator model has choices: the candidate reference subspace $\mathcal{R}_t$, the gate, the selector, the sense-subspace dimension, and the spectra in Eq. (30). The eigenvalues are not identified by the subspaces alone. These choices must be fixed or learned under held-out evaluation; otherwise the framework risks describing whatever geometry is found after the fact.

D. Empirical predictions

The key empirical task is to separate the geometric bracket from the stronger reference-subspace and intertwined-observable claims.

Claim G: a calibrated Goldilocks regime. For pre-specified ε, δ , and τ , useful overcomplete representations should lie above the packing and readout lower bounds in Eq. (21), while retaining $N > d_{\text{eff}}$.

Claim A: privileged token reference subspace. For a polysemous token t , there exists a privileged token-associated subspace $\mathcal{R}_t$ such that sense-relevant variation is better exposed in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces of the same dimension.

Claim B: low-dimensional sense subspaces. Within $\mathcal{R}_t^\perp$, sense-conditioned states concentrate near low-dimensional subspaces with small cross-sense coherence.

Claim C: intertwined context operators. Context projectors $\Pi_{t,C}$ share P_t , differ on $\mathcal{R}_t^\perp$, and improve held-out contextual prediction or compression relative to dimension-matched unstructured projectors. Where contexts are operationally incompatible, their fitted commutator should be nonzero and sequential projection should be order-sensitive.

These claims can fail independently. Claim G concerns the global dictionary and active-set geometry. Claim A can hold while sense information remains high-dimensional. Claim B can hold relative to some non-token-associated subspace, in which case Claim A fails. Claims A and B can hold even if the operator constraint in Claim C adds no predictive value.

The following tests separate the possibilities.

1. *Goldilocks calibration test.* Estimate d_{eff} by a declared participation-ratio or intrinsic-dimension estimator, count a prespecified dictionary of N feature directions, measure its maximum overlap, and estimate active-set size k and tolerated readout noise τ . Compare observed failure and recovery rates with Eqs. (16) and (21). The comparison must include isotropic random dictionaries and trained, coactivation-matched controls.
2. *Reference-subspace test.* For a polysemous token, choose candidate reference subspaces generated by the normalized static input embedding, mean contextual embedding, learned token-specific directions, or a low-dimensional combination of these. Project layer states as in Eq. (41). Sense labels should be better separated in $\mathcal{R}_t^\perp$ than in orthogonal complements of random, frequency-matched, or layer-matched control subspaces of the same dimension. The statistic may be classification accuracy or mutual information with sense labels, assessed by permutation over sense labels and bootstrap over token instances.
3. *Operational subspace test.* For each sense s , collect projected states $z_\ell(t, x) \in \mathcal{R}_t^\perp$ and estimate a low-rank subspace by PCA or probabilistic factor analysis. With $U_{t,s}$ the first q principal directions for sense s , measure cross-sense coherence $\|U_{t,s}^\top U_{t,s'}\|_{\text{op}}$. True sense labels should yield lower cross-sense coherence than shuffled labels and stronger separation than matched control reference subspaces.
4. *Gated-readout test.* Linear probes trained on $z_\ell(t, x)$ should perform comparably to probes trained on the full state $h_\ell(t | x)$, while probes restricted to the reference component $P_t h_\ell(t | x)$ should perform near chance on sense classification.
5. *Intertwined-observable test.* Fit P_t and $B_{t,C}$ on training contexts, construct $\Pi_{t,C}$ and $A_{t,C}$, and evaluate Eq. (31) on held-out instances. Compare against unconstrained low-rank projectors, mixture-of-experts probes with the same parameter count, and shuffled context labels. Report $\|[\Pi_{t,C}, \Pi_{t,C'}]\|_{\text{op}}$ and test whether reversing sequential context projections changes predictions where the model claims contextual incompatibility.
6. *Overlap-distribution test.* If a representation exploits quasi-dimensionality, pairwise inner products among token embeddings or SAE feature directions should concentrate near zero with variance $\sim 1/d_{\text{eff}}$. Deviations quantify anisotropy and effective dimension loss.
7. *Synthetic concurrency degradation.* At a fixed layer, inject controlled superpositions

$$h_k = h_0 + \lambda \sum_{j=1}^k x_j w_j, \quad (48)$$
 using directions w_j obtained independently, for example from SAE decoder directions or frozen linear probes. As k increases, readout should degrade smoothly on the scale $\sqrt{k/d_{\text{eff}}}$ for random-like directions, with weaker degradation for feature sets that training has orthogonalized.
8. *Coactivation-dependent geometry.* Frequently coactive features should show reduced mutual coherence relative to rarely coactive features. This tests the broader capacity/accessibility framework through Eq. (47), independently of the reference-subspace hypothesis.

E. Scope of the quantum analogy

The analogy to quantum contextuality is structural and limited. Both settings can describe alternatives by observables with context-dependent eigenspaces. In the proposed construction, the operators $A_{t,C}$ intertwine because they share the spectral projector P_t ; their complementary sense sectors may yield nonzero commutators. The hyperedges in Fig. 2 record only this incidence pattern, whereas Eqs. (28)–(30) supply its operator realization. A basis is a set of vectors, a subspace is their span, and an observable is an operator with a spectral decomposition; these objects are not interchangeable.

Quantum contextuality has the force of no-go theorems such as Kochen–Specker [32, 33]; no analogous theorem is claimed here for semantic embeddings. Real symmetric matrices can be noncommuting while still being stored and evaluated on an ordinary classical computer. Thus the operator construction is quantum-inspired, but non-commutativity by itself supplies neither Born probabilities nor physical quantumness.

Three levels of analysis should not be conflated.

1. *Classical high-dimensional representation.* A present-day Transformer carries real-valued hidden states through attention, nonlinear feed-forward maps, residual connections, and normalization. Concentration, quasi-orthogonality, feature superposition, dynamic embeddings, and finite readout are all available at this level. None requires quantum probability or quantum hardware.
2. *Quantum-like semantic or contextual formalism.* States, contexts, bases, interference terms, non-commuting questions, or measurement-like updates may be used as an abstract model of meaning even when the implementation remains classical. The relevant issue is not the vocabulary but whether this extra structure improves compression, explanation, or prediction relative to an ordinary contextual embedding model.
3. *Genuine quantum machine learning or quantum attention.* Here semantic states would be encoded in qubits or other physical quantum degrees of freedom, intermediate evolution would be unitary, phase would affect interference, and measurement would follow quantum probability. A standard Transformer does none of these things.

The Goldilocks bracket belongs entirely to the first level: it is a claim about classical high-dimensional geometry. The shared-subspace hypothesis is likewise testable in a classical model. The intertwined operators and their order-sensitive projections belong to the second level because they make the quantum-inspired comparison explicit and implementable. Their scientific value depends on the discriminative test in Sec. VD: they must improve prediction, compression, or explanation relative to equally expressive classical baselines. Even a positive result would support a useful quantum-like formalism, not physical quantum computation; phase-sensitive dynamics and Born-rule measurement are still absent.

The dimension of $\mathcal{R}_t$ is also explicit. In Hilbert-space quantum logic, different contexts can share one observable, as in the familiar tripod-like case, or several observables; a four-dimensional example has two contexts with two common observables [13]. The semantic analogue is $\dim(\mathcal{R}_t) = p$: $p = 1$ is the simplest case, while $p > 1$ permits several token-associated reference directions. In every case, sense-specific information is hypothesized to occupy subspaces of $\mathcal{R}_t^\perp$.

VI. CONCLUSION

Useful quasi-dimensionality is not obtained by making a representation merely “large.” It occupies a bracket. Too few represented directions relative to d_{eff} leave an exact orthogonal code available; too small an effective carrier for a fixed dictionary produces excessive pairwise overlap; and too many jointly active directions produce

accumulated readout noise. Once the tolerances are specified, the operative statement is

$$d_{\min} = \max \left\{ d_{\text{pack}}(N, \varepsilon, \delta), \frac{k}{\tau^2} \right\} \lesssim d_{\text{eff}} < N. \quad (49)$$

The exact packing term follows from the spherical-cap probability in Eq. (15); the readout term follows from $\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}$. This is the quantitative distinction between kinetic semantic capacity and epistemic semantic accessibility.

There is no context-free universal value of the “right dimension.” The answer depends logarithmically on dictionary size N , quadratically on the allowed overlap ε , linearly on the active-set size k , and quadratically on the inverse readout tolerance $1/\tau$. Under the illustrative choices $N = 10^4$, $\varepsilon = 0.10$, $\delta = 0.05$, $k = 64$, and $\tau = 0.125$, the calculation gives $4.1 \times 10^3 \lesssim d_{\text{eff}} < 10^4$. The value is not offered as a universal LLM constant; it demonstrates how an empirical study can compute its own Goldilocks zone rather than appeal to dimension qualitatively.

The second central construction implements intertwined contexts. For a polysemous token, the context observables $A_{t,C}$ share the spectral projector P_t associated with token identity and differ on sense subspaces of $\mathcal{R}_t^\perp$. Their projectors $\Pi_{t,C}$ can fail to commute, and conditional projection $h_{t,C} = \Pi_{t,C}h$ supplies a context-specific embedding. A soft selector forms the mixture in Eq. (45). The “bank” schematic displays the financial, river-edge, and seat/bench contexts; the aircraft-banking sense is a possible fourth hyperedge rather than a replacement for the intended seat association.

These two ideas have different evidential status. The Goldilocks bracket is a classical geometric null model whose parameters can be measured directly. The shared reference subspace, the noncommuting context operators, and their predictive usefulness are stronger hypotheses. They must outperform dimension- and parameter-matched classical baselines on held-out data. A failure of the operator model would leave the capacity/interference bracket intact; a failure of the random-code bracket would reveal learned structure, anisotropy, or nonlinear readout not captured by the null model.

Finally, the operator construction is quantum-inspired, not a claim that a Transformer is a quantum device. It uses real matrices on classical hardware and introduces no Born rule, unitary dynamics, or physical measurement. Its value lies in whether shared spectral projectors, incompatible context operators, and order-sensitive conditional embeddings provide a more compact or predictive account of polysemy. That question is discriminative and falsifiable; the vocabulary of quantum contextuality is useful only to the extent that the proposed tests answer it positively.

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